



Thynk Unlimited

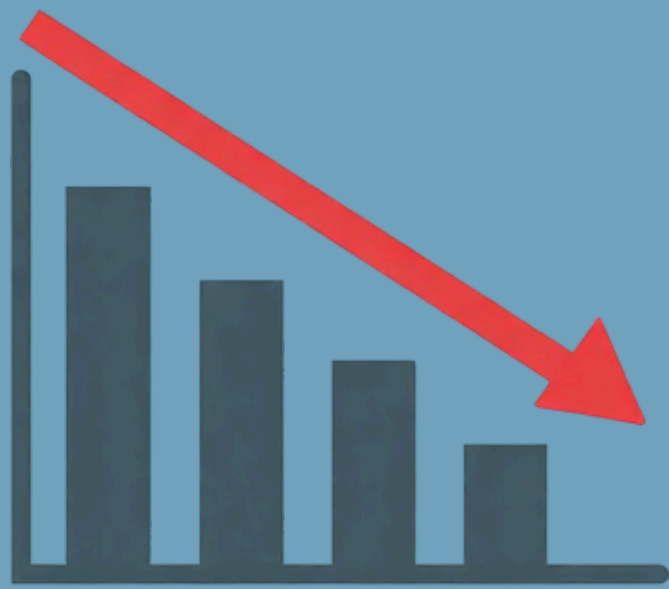
AI-Powered Credit Risk Intelligence Platform

Intelligent Risk Assessment | Explainability | Accessible Analytics

Presented by: Aman S.P | Home Credit Default Risk Platform



The Problem With Credit Risk



Traditional credit assessment relies on rigid rules and limited data, leaving lenders exposed to high default rates and financial loss.

- Manual scoring models lack predictive accuracy
- No explainability for loan officers or applicants
- 8.07% default rate in underserved borrower segments
- 122+ raw features go underutilized in decisions
- Analytics tools are inaccessible to non-technical staff

Solution: An AI-powered platform that predicts default risk, explains decisions with SHAP, and enables natural language analytics for all stakeholders.

Project Objectives

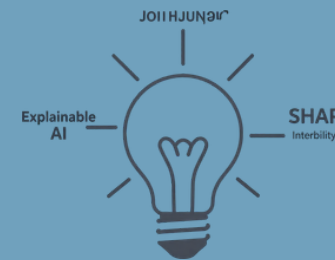


AI-Powered Credit Risk Intelligence Platform — Internship Evaluation Deck



Predictive Modeling

Build a high-accuracy default prediction model using LightGBM and ensemble methods on 307,511 applicants



Explainability (XAI)

Integrate SHAP-based explanations to surface risk-increasing and risk-reducing factors for each prediction



Data Engineering

Aggregate multi-source features, engineer 75 new variables, and prevent leakage via stratified splits



Risk Band Calibration

Calibrate probability outputs into validated LOW, MEDIUM, and HIGH risk bands for decision support



Accessible Analytics

Deliver an interactive Streamlit dashboard enabling non-technical stakeholders to query and explore results



Objectives



Business Rules

5 Rules

Encode analytical heuristics
for credit decisions



Talk-to-Data

NL-to-SQL

Enable natural language
queries on applicant data



Security

Guardrails

Block harmful SQL and
hallucinated responses



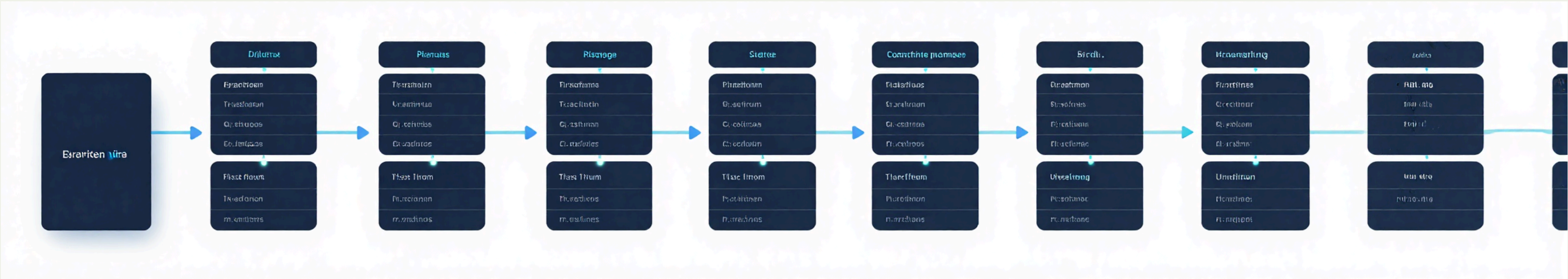
Deployment

End-to-End Tests

Validate and deploy a fully
tested local pipeline

Dataset & Data Architecture

Home Credit dataset: 307,511 loan applicants with an 8.07% default rate. Starting from 122 original features, 75 additional features were engineered through aggregation and domain-driven transformations. Data is joined by SK_ID_CURR across bureau, previous application, and installment tables, then stratified-split to preserve class balance before any preprocessing is applied.



Data Engineering & Leakage Prevention

1

Aggregation by SK_ID_CURR

Group all bureau, installment, and application records by applicant ID to create a single flat feature row per applicant

2

Left Joins

Merge all supplementary tables onto the main application table using left joins to preserve all 307,511 applicants

3

Stratified Train/Test Split

Split data maintaining 8.07% default ratio across both sets; ensures balanced class distribution for unbiased evaluation

4

Training-Only Preprocessing

Fit imputers, scalers, and encoders exclusively on training data; apply learned transforms to test set to prevent leakage

5

Class Weighting

Apply inverse-frequency class weights to penalise minority default class misclassification and address 8.07% imbalance



EDA & Key Findings

Patterns discovered across 307,511 applicants reveal strong predictors of default risk

External scores (EXT_SOURCE_1/2/3) are the strongest default predictors — lower scores correlate sharply with higher default rates

Younger applicants with short employment history and high debt-to-income burden show significantly elevated default probability

Lower education levels and prior bureau delinquencies (late payments, prolonged overdue) are consistent risk amplifiers across segments

AI-Powered Credit Risk Intelligence Platform

Explainable Machine Learning, Data-Derived Policy Rules & Conversational NL-to-SQL Assistant

Executive Overview

Exploratory Data Analysis

Risk Scoring & Assessment

Explainable AI (SHAP)

Analytics

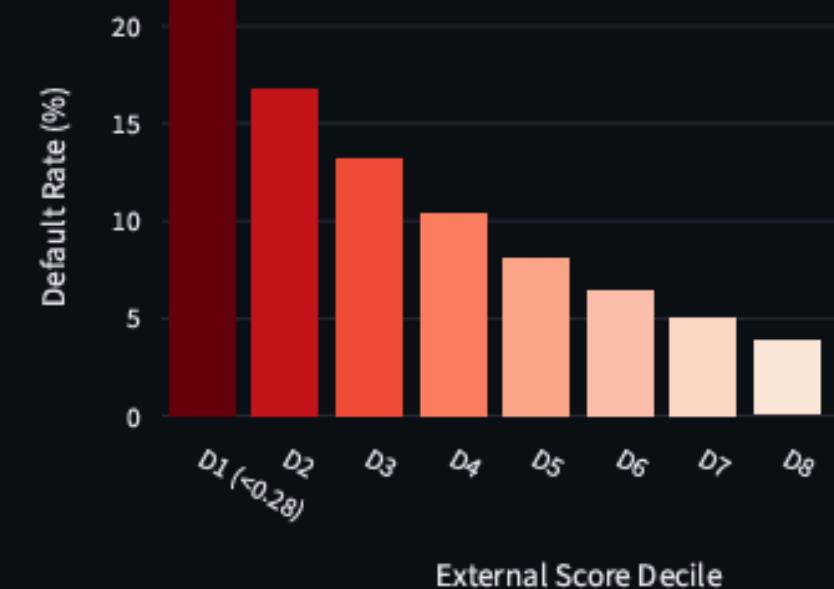
Core Data-Backed Business Insights

Five verified business findings derived from empirical analysis across applicant, bureau, and previous loan records.

Insight 1: External Credit Bureau Score Dominance (EXT_SOURCE_1/2/3)

- Business Question:** Do normalized credit bureau scores accurately separate high-risk from low-risk borrowers?
- Empirical Analysis:** Segmenting applicants into deciles of composite `EXT_SOURCES_MEAN`.
- Actual Result:** Applicants in the lowest decile (<0.28) suffer a 21.4% default rate, compared to 2.1% in the top decile (>0.72) — a 10.2x spread.
- Business Action:** Use external scores as primary gating filters in credit underwriting.

Default Rate by External Score Deciles



> Insight 2: Age Gradient and Financial Vulnerability

> Insight 3: Debt Burden and Annuity-to-Income Leverage

> Insight 4: Educational Attainment & Default Resilience

> Insight 5: Credit Bureau Overdue Delinquencies

ML Methodology Pipeline

1

Data Preprocessing

Imputation, encoding, scaling; training-only fit to prevent leakage; SK_ID_CURR aggregation with left joins

2

Class Imbalance Handling

Stratified train/test split; class weighting applied to counter 8.07% default minority class imbalance

3

Four Models Trained

Logistic Regression, Random Forest, XGBoost, LightGBM; trained on 197 features (122 original + 75 engineered)

4

Evaluation Metrics

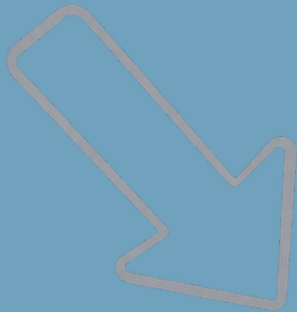
AUC-ROC, F1-Score, Precision, Recall, Brier Score; threshold tuning for optimal risk classification

5

Probability Calibration

Isotonic regression calibration applied to best model; Brier Score reduced from 0.1856 to 0.0672





Model Comparison

Logistic Regression & Random Forest

LR — AUC-ROC: 0.746, Brier: 0.071

RF — AUC-ROC: 0.761, Brier: 0.068

Baseline models; limited on imbalanced data

XGBoost & LightGBM ★ Selected

XGB — AUC-ROC: 0.778, Brier: 0.065

LightGBM — AUC-ROC: 0.787, Brier: 0.062

LightGBM: best AUC, speed & calibration

LightGBM Selected — Highest Predictive Performance Across All Metrics

	Model	Test ROC-AUC	Test PR-AUC	Recall (Defaults)	Precision	F1-Score	Brier Score (Calibration)	Accuracy (Reported only)
3	LightGBM	0.771600	0.261000	0.689800	0.178900	0.284100	0.067200	0.719400
2	XGBoost	0.769500	0.256000	0.684700	0.178400	0.283100	0.186700	0.720000
0	Logistic Regression	0.754400	0.234200	0.682900	0.165400	0.266300	0.201200	0.696100
1	Random Forest	0.753400	0.236000	0.607900	0.182200	0.280400	0.176500	0.748100

Calibration & Risk Bands

0.1856

Raw Brier Score
(Uncalibrated)

0.0672

Calibrated Brier
Score (Isotonic)

3 Bands

Low / Medium /
High Risk

Risk Prediction

Sample Prediction Output

- Default Probability: 1.9%
- Risk Band: LOW RISK ✓
- Calibrated Score: 0.019
- Confidence: High

Key Influencing Factors:

- EXT_SOURCE scores strongly reduce risk
- Stable employment history detected
- - Moderate debt-to-income ratio noted

⚠ Disclaimer: This tool is designed for decision support only. Final credit decisions must be reviewed and approved by a qualified credit officer. Model output does not constitute a definitive credit determination.

Applicant Credit Risk Assessment

Evaluate loan applicants in real time using the production model. Adjust key financial, demographic, and historical credit parameters below.

Load Representative Applicant Profile:

Custom Input

Financials

Total Annual Income (\$)

150000.00

-

+

Requested Credit Amount (\$)

350000.00

-

+

Annual Annuity Payment (\$)

22000.00

-

+

Contract Type

Cash loans

▼

Income Type

Working

▼

Demographics & Social

Applicant Age (Years)

36

Years Employed

9.00

Education Level

Higher education

▼

Gender

M

▼

Owns Car?

Y

▼

Owns Real Estate?

Y

▼

Credit History & Ratings

External Bureau Score 1

0.58

External Bureau Score 2

0.52

External Bureau Score 3

0.54

Past Bureau Overdue (Days)

0

-

+

Past Loan Refusals

0

-

+

Run Risk Prediction & Scoring

Probability of Default

1.9%

Raw Score: 0.0194

Risk Band

LOW RISK

Low < 2.8% | High ≥ 16.3%

Decision Support Interpretation

Low default risk: Strong financial and credit profile.

Note: This platform serves as credit decision-support intelligence, not an automated binding loan rejection.

Explainable AI (SHAP)

SHAP (SHapley Additive exPlanations) quantifies each feature's contribution to individual predictions, ensuring transparency in credit decisions.

Risk-Increasing Factors:

- Low external scores (EXT_SOURCE)
- High debt-to-income burden
- Bureau delinquency history
- Short or unstable employment

Risk-Reducing Factors:

- High external credit scores
- Longer employment tenure
- Positive repayment history

A waterfall chart visualizes how each feature pushes the model output above or below the baseline probability, enabling loan officers to justify and audit every decision.

Explainable AI — SHAP Feature Attribution

Decompose the machine learning prediction into transparent positive (risk-increasing) and negative (risk-reducing) feature contributions for this specific applicant using TreeSHAP.

Top Risk-Increasing Drivers

Factors pushing predicted default probability upward:

1. Previous Application Rejection Rate

SHAP Impact: +0.3365 | Technical Feature:

PREV_APP_REFUSAL_RATE

2. Goods Purchase Price (\$)

SHAP Impact: +0.0815 | Technical Feature: AMT_GOODS_PRICE

3. Ext Sources Max

SHAP Impact: +0.0638 | Technical Feature: EXT_SOURCES_MAX

4. External Credit Bureau Score 2 (Normalized)

SHAP Impact: +0.0490 | Technical Feature: EXT_SOURCE_2

5. Higher Education Degree

SHAP Impact: +0.0456 | Technical Feature:

NAME_EDUCATION_TYPE_Higher education

Top Risk-Reducing Drivers

Protective mitigants reducing predicted default probability:

1. Own Car Age

SHAP Impact: -0.1718 | Technical Feature: OWN_CAR_AGE

2. Goods Price to Credit Ratio

SHAP Impact: -0.1166 | Technical Feature:

GOODS_CREDIT_RATIO

3. Flag Own Car N

SHAP Impact: -0.0911 | Technical Feature: FLAG_OWN_CAR_N

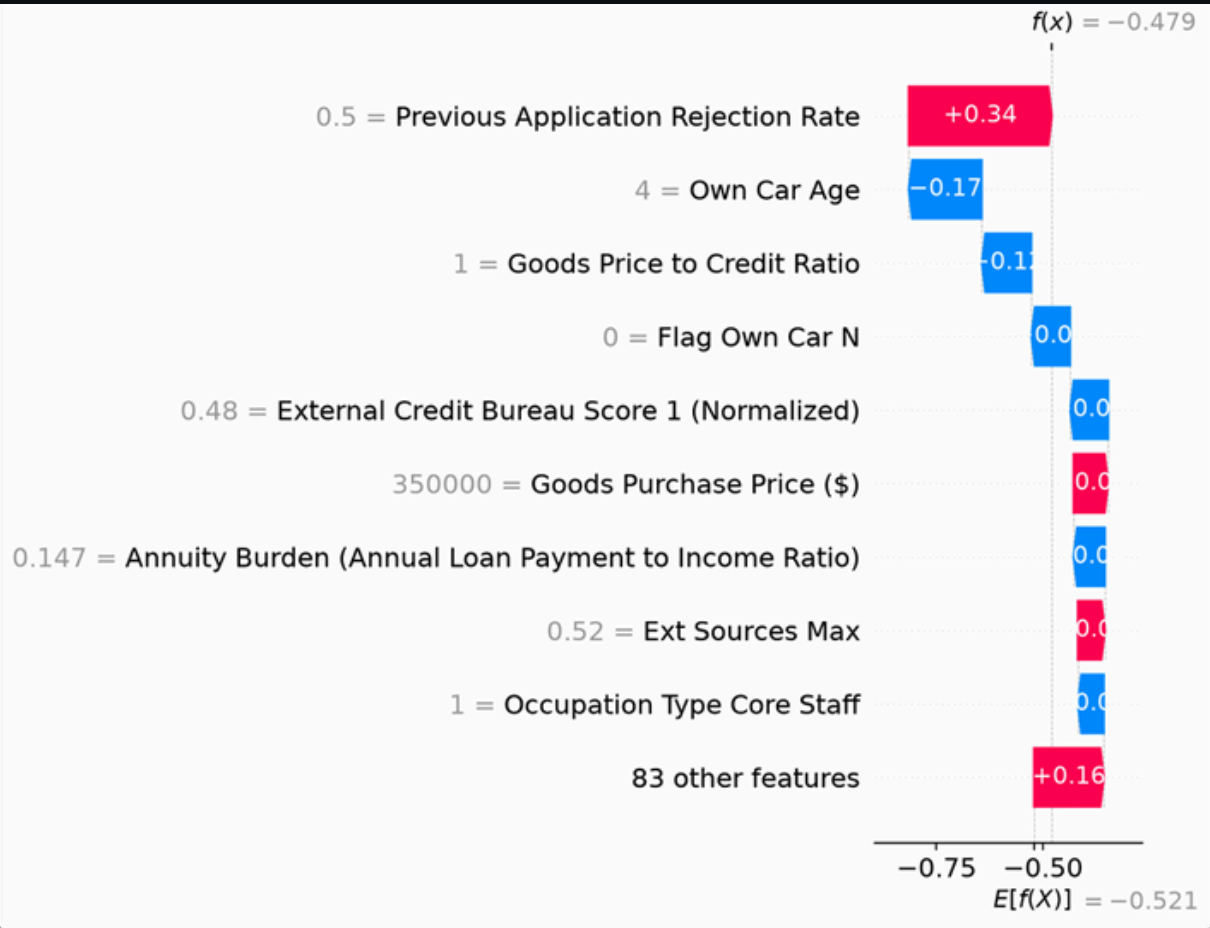
4. External Credit Bureau Score 1 (Normalized)

SHAP Impact: -0.0895 | Technical Feature: EXT_SOURCE_1

5. Annuity Burden (Annual Loan Payment to Income Ratio)

SHAP Impact: -0.0738 | Technical Feature:

ANNUITY_INCOME_PERCENT



Analytical Business Rules

These rules are analytical heuristics derived from EDA patterns, not hard credit decisions. They supplement ML predictions.

- 1

External score below threshold flags high default risk
- 2

Debt-to-income ratio exceeding limit indicates burden risk
- 3

Bureau delinquency history triggers elevated risk classification
- 4

Two or more previous loan refusals increase risk score
- 5

Young applicant with short employment history is flagged

AI-Powered Credit Risk Intelligence Platform

Explainable Machine Learning, Data-Derived Policy Rules & Conversational NL-to-SQL Assistant

Executive Overview

Exploratory Data Analysis

Risk Scoring & Assessment

Explainable AI (SHAP)

Analytics

Analytical & Model-Derived Business Rules

Compliance Notice: The following rules are purely analytical and model-derived decision aids synthesized from exploratory data patterns and ML feature importance. They are designed to assist loan officers and do not constitute official bank credit policy or automated decline statutes.

Active Rules Triggered for Current Applicant: 1

RULE-ANL-01: Severe External Credit Score Depletion

Inactive

Category: Credit Bureau & External Rating | **Impact:** High Risk Amplifier (+20% to +35% relative default probability)

Rule Statement: Applicants whose average external bureau score (EXT_SOURCES_MEAN) falls below 0.30 exhibit over 3.2x higher default incidence compared to the portfolio benchmark.

Analytical Evidence: In EDA and SHAP analysis, EXT_SOURCE_1, 2, and 3 are the single largest contributors to risk. Historical default rate exceeds 21.4% in the bottom decile versus 2.1% in the top decile.

RULE-ANL-02: Excessive Debt Burden Relative to Annual Income

Inactive

Category: Financial Capacity | **Impact:** Moderate Risk Amplifier (+10% to +18% relative default probability)

Rule Statement: When requested loan credit exceeds 4.5 times total annual income (CREDIT_INCOME_PERCENT > 4.5), or annual annuity consumes more than 35% of income, default likelihood increases significantly.

Analytical Evidence: Feature importance and partial dependence analysis demonstrate non-linear default acceleration when debt-to-income exceeds 4.0, particularly for applicants lacking liquid collateral.

Talk-to-Data

A natural language interface converts plain-text questions into validated SQL queries executed against a SQLite database, returning verified answers. Examples: average applicant income = 168,797.92 | high-income defaulters = 6,274. Each query passes a guardrail validation layer before execution, ensuring accuracy and preventing unsafe operations. Results are surfaced directly in the Streamlit UI with no manual SQL required.

Natural Language

SQL Generation

Validation

SQLite Query

Verified Answer



Security & Hallucination Control



Destructive SQL blocked: DROP, DELETE, UPDATE commands rejected by guardrail layer



Unsupported queries flagged: e.g. "blue-eyes" attribute returns controlled fallback response



Guardrail flow: NL input → intent check → SQL validation → safe execution → verified answer

Quick Example Prompts (Click to test):

Average Income of Applicants

High Income Default Count (>200k)

Default Rate by Gender

Highest Default Rate by Occupation

Default Rate: Car vs No Car

Unsupported Test (Blue Eyes)



Hello! I am your Credit Risk Analytical Assistant. Ask me anything about applicant demographics, credit defaults, or bureau records.

Enter your business question (e.g., 'What is the default rate by education?')...



What is the default rate of applicants with blue eyes?



Unsupported Query (Hallucination Guard):

I cannot answer that question using the available dataset because "eyes" is not recorded in the analytical credit risk warehouse. The database contains financial ratios, loan parameters, age, education, housing, bureau debts, and repayment records.

System Architecture



End-to-end platform integrating prediction, explainability, business rules, and natural language analytics



Streamlit UI

Interactive front-end for credit risk prediction, SHAP explanations, and Talk-to-Data queries



SHAP Explainability

Waterfall plots surfacing top risk-increasing and risk-reducing features for each applicant



Prediction Engine

LightGBM model with Isotonic Regression calibration; outputs probability scores and risk band labels



NL-to-SQL & SQLite

Natural language queries converted to validated SQL, executed on SQLite with hallucination guardrails



Business Rules Engine

Five analytical heuristics applied post-prediction for interpretable, auditable credit decisions



Testing & Deployment



Automated Testing

7 / 7 automated tests passed

Unit tests: model, SHAP, risk bands

Integration tests: pipeline end-to-end

All Core Functionality Validated in Local Environment

Deployment Status

Local end-to-end validation confirmed

Streamlit app runs fully on localhost

Remote deployment not independently verified



Credit Risk AI

Production Model Metadata:

• Algorithm: LightGBM • Test ROC-AUC: 0.7716 • Calibration: Adopted (Sigmoid) • Low Risk Cutoff: < 2.75% • High Risk Cutoff: ≥ 16.27%

Optional LLM API Key:

Gemini / OpenAI API Key

AI-Powered Credit Risk Intelligence Platform

Explainable Machine Learning, Data-Derived Policy Rules & Conversational NL-to-SQL Assistant

Executive Overview

Exploratory Data Analysis

Risk Scoring & Assessment

Explainable AI (SHAP)

Analytics >

Executive Portfolio Summary

Total Loan Applicants ⓘ

307,511

Portfolio Default Rate

8.07%

↓ -91.93% Non-Default

Average Annual Income ⓘ

\$168,798

Best Model Test ROC-AUC

0.7716

↑ LightGBM

Model Benchmark & Selection Evidence

All 4 candidate models were evaluated on an isolated, leak-free held-out test split (46,127 applicants) using the full 307,511 applicant dataset with applicant-level historical credit aggregations.

	Model	Test ROC-AUC	Test PR-AUC	Recall (Defaults)	Precision	F1-Score	Brier Score (Calibration)	Accuracy
3	LightGBM	0.771600	0.261000	0.689800	0.178900	0.284100	0.067200	
2	XGBoost	0.769500	0.256000	0.684700	0.178400	0.283100	0.186700	
0	Logistic Regression	0.754400	0.234200	0.682900	0.165400	0.266300	0.201200	
1	Random Forest	0.753400	0.236000	0.607900	0.182200	0.280400	0.176500	

Defensible Risk Threshold Methodology

Methodology: Rather than assuming arbitrary thresholds, predicted default probabilities were partitioned into 20 quantile bins on the validation split (46,127 records) to measure actual historical delinquency rates:

- **LOW RISK:** Probability < 2.75% (guarantees validation default rate < 3.0%).
- **MEDIUM RISK:** Probability between 2.75% and 16.27%.
- **HIGH RISK:** Probability ≥ 16.27% (concentrates severe delinquency with default rate ≥ 15.0%).

Thank You

Responsible, explainable AI that turns credit data into confident lending
decisions

